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IFUL: A Python package for joint-modeling strong gravitational lensing and their source kinematics

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Summary

IFUL (Integral Field Unit Lensing) is an open-source Python package for joint modeling and simulating strong gravitational lensing systems alongside the spatially resolved internal kinematics of background source galaxies. Strong gravitational lensing magnifies and distorts light from a background galaxy (the "source") around a massive foreground object (the "lens") into multiple images and/or arcs. This phenomenon serves as a primary probe in observational astrophysics for measuring cosmological parameters (e.g., the Hubble constant H_0, the dark energy equation of state w, and the matter density $\Omega_{\mathrm{m},0}$), testing dark matter models, studying high-redshift galaxy evolution across cosmic time, and many more applications.

While traditional lens modeling tools rely primarily on 2D photometric imaging (x and y axes), IFUL incorporates 3D Integral Field Unit (IFU) spectroscopic datacubes (x, y, and λ axes) into an end-to-end forward-modeling framework. Integral field spectroscopy provides a full spectrum for every spatial pixel (spaxel), delivering spatially resolved measurements of stellar and gas kinematics. This allows our framework to incorporate not only spatial flux but also dynamical information into the lens model. IFUL forward-models every spaxel in the 3D datacube by unifying macro lens mass profiles with physical models of source galaxy kinematics. By leveraging spatially resolved kinematic markers within lensed arcs and isolating source line emission from foreground lens light, IFUL yields competitive constraints on both lens mass distributions than high-resolution photometric imaging data alone, while also providing insight into high-redshift source dynamics.

Statement of need

Strong gravitational lens modeling is currently in a regime where precision cosmography and mass profile determinations are dominated by systematic modeling uncertainties rather than statistical noise. Traditional lens modeling pipelines rely on 2D photometric imaging. While 3D IFU spectroscopy (from space and ground-based observatories such as *JWST* (NIRSpec, MIRI), VLT (MUSE, ERIS), Keck (KCWI/KCRM), and ALMA) is routinely acquired for lensing fields, these datacubes are typically underutilized. In conventional workflows, IFU data are restricted to measuring integrated deflector stellar velocity dispersions or confirming source galaxy redshifts, while overlooking the rich spectral and spatial information contained in individual spaxels of the source galaxy.

IFUL advances lens modeling beyond traditional 2D imaging through several key capabilities:

- Kinematic markers as constraints: When a source galaxy exhibits coherent rotation and/or dynamic structure, velocity gradients (both line-of-sign velocities v_{los} and velocity dispersions σ_v) serve as spatial markers across multiple lensed images, providing powerful complementary constraints on the lens mass profile.

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- Continuum subtraction and image disambiguation: By performing continuum subtraction across the 3D datacube, `IFUL` isolates narrow emission lines from the background source galaxy. This removes severe flux contamination from the bright foreground lens, unblending overlapping components and revealing faint counter-images otherwise buried under lens continuum light.
- Intrinsic source kinematic reconstruction: Concurrent forward-modeling of the macro lens and 3D IFU datacube synthesizes spaxel data from all lensed images into a unified source-plane model. This enables robust determination of the source galaxy's intrinsic line-of-sight velocity and velocity dispersion fields while fully marginalizing over lens-model parameters.
- Systematic uncertainties: Being an alternative to traditional 2D modeling, `IFUL` offers an independent lens model sourced from datacubes. This serves as an additional probe for systematic uncertainties in a regime of overly-constrained lens models.

Despite the growing availability of high-resolution IFU observations from current facilities and upcoming IFU instruments (such as ELT/HARMONI, ELT/MICADO, and TMT/IRIS), no existing open-source software was dedicated to full 3D forward-modeling of lensed IFU datacubes. Individual research groups were forced to write custom, non-standardized codebases from scratch, creating barriers to entry and hindering scientific reproducibility. `IFUL` fills this software gap by delivering an open-source, modular Python package built to interface with established lens modeling tools like `lenstronomy` (Birrer et al., 2021; Birrer & Amara, 2018).

`IFUL` is designed for astronomers, astrophysicists, and observational cosmologists working on strong gravitational lensing, high-redshift galaxy kinematics, and 3D spectroscopic analysis.

State of the field

Strong gravitational lens mass modeling is supported by a mature ecosystem of open-source software packages. Widely adopted 2D Python packages include `lenstronomy` (Birrer et al., 2021; Birrer & Amara, 2018), `PyAutoLens` (Nightingale et al., 2021), `glafic` (Oguri, 2010), `gigalens` (Gu et al., 2022), `gravlens` (Keeton, 2001), and `GLEE` (Suyu et al., 2012; Suyu & Halkola, 2010). While these frameworks excel at modeling high-resolution photometric images (e.g., from *HST* or *JWST*), they operate exclusively on 2D spatial pixel arrays. As such, existing tools lack the data structures and forward-modeling engines necessary to process 3D IFU spectroscopic datacubes.

Although the scientific benefit of combining source kinematic reconstruction with strong gravitational lensing is recognized, existing 3D approaches remain limited:

- Early 3D implementations: Bolton & Burles (2007) developed an early 3D IFU lens modeling code in IDL, but it was proprietary, unreleased, and predated modern high-resolution spectrographs (e.g., VLT/MUSE, Keck/KCWI, *JWST*/NIRSpec).
- Proprietary Bayesian 3D methodologies: Rizzo et al. (2018) introduced a 3D modeling technique designed to recover lensed source kinematics. However, their underlying codebase remains private, and their method was demonstrated only on simulated data with no published application to observational data.
- Decoupled two-stage kinematic mapping: Workflows such as Chirivì et al. (2020) and Zhou et al. (2026) extract 2D kinematic maps ($v_{\rm los}$, σ_v) in the image plane before ray-tracing them back to the source plane. While useful, this decoupled approach avoids full 3D forward-modeling and suffers from systematic errors caused by spatial blending, beam smearing, continuum contamination, and lost spaxel-level covariances. Furthermore, no standardized open-source software package exists for this two-stage methodology.

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IFUL provides the only open-source solution dedicated to 3D IFU datacube lens modeling. While built with lenstronomy as its underlying 2D spatial solver, IFUL was developed as a dedicated package because 3D datacube forward-modeling requires distinct data structures, source kinematic profiles, and a 3D reconstruction framework.

Software design

A core architectural principle of IFUL is modularity. Rather than re-implementing 2D gravitational ray-tracing models, IFUL uses lenstronomy as its underlying 2D spatial solver engine. This choice grants IFUL immediate access to lenstronomy's comprehensive catalog of mass profiles (e.g., EPL, NFW, SIS, external shear) while focusing its own scope on organizing 2D spatial image planes into a unified 3D physical forward-modeling framework. The same can be said about how we handle the kinematic mapping of the source galaxy. Powerbin (Cappellari, 2025) has already a sophifisticated and scientifically-proven (Sheu et al., 2026) methodology in discretizing spaxel regions based on their signal-to-noise. As such, we utilize Powerbin for our source kinematic reconstruction rather than reinvent the wheel.

Building off our principle of modularity, IFUL enforces a strict separation of concerns across its pipeline. Datacube ingestion, spaxel continuum background subtraction, spatial masking, and noise estimation are encapsulated within the ImageSet module, completely isolated from lensing physics. Spatial coordinate transformations and 2D ray-tracing abstractions are managed by FlatModel, while 3D spatial-spectral synthesis and kinematic model evaluations are orchestrated by IFULModel. Decoupling data preprocessing, 2D ray-tracing, physical kinematic modeling, and simulation tools allows researchers to easily integrate custom velocity profiles, alternative continuum subtraction routines, or adaptive spatial binning methods without altering the underlying likelihood or lens-solver engines.

Research impact statement

IFUL has already demonstrated its scientific utility through direct application to observational datasets. In an upcoming study (Sheu et al. in prep.), IFUL was deployed to perform joint 3D spatial-spectral lens modeling of the "Carousel" strong gravitational lens system using high-resolution VLT/MUSE IFU observations (O'Donnell et al., 2026; Sheu et al., 2024; Urcelay et al., 2026). By isolating emission line kinematics from foreground lens continuum light, IFUL successfully reconstructed the intrinsic line-of-sight velocity field and velocity dispersion of the background spiral galaxy at redshift $z = 1.432$. Crucially, incorporating spaxel-level spectroscopic constraints yielded tighter posterior probability distributions on macro lens mass model parameters compared to traditional 2D photometric modeling alone, while enabling robust constraints on high-redshift disk dynamics and central supermassive black hole mass limits.

Beyond individual lens system modeling, IFUL has been presented at international astrophysics conferences, receiving strong interest from the gravitational lensing and galaxy evolution communities. The software is currently being adopted by multiple research groups for ongoing projects, including the 3D spectroscopic modeling of lensed quasar host galaxies and lensed star-forming systems observed with *JWST*/NIRSpec. For many strong lensing groups, the IFU data already exists (e.g., Knabel et al., 2025; Lin et al., 2026). IFUL is already planned to be used for further refinement to the lens model of these systems, and as an additional probe to study source kinematics in future analyses.

AI usage disclosure

Generative AI (Gemini 3.6 Flash) was used to check formatting, spelling, and grammar of the manuscript, as well as to assist in generating docstrings for the codebase. No generative AI

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was used in the creation of the underlying code itself.

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